

Lab04_ME315

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```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
fname <- "\\smb\ra281972\Documentos\WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

```
countries <- read_excel(fname, sheet = "Country")
```

```
países_selecionados <- c("BRA", "ARG", "CHL", "COL", "PER")
countries <- countries |>
  filter(`Country Code` %in% países_selecionados)
```

```
WDI <- read_excel(fname, sheet = "Data")
```

```
WDI <- WDI |>
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
```

```
WDI_tidy <- WDI |>
  pivot_longer(cols = `1960`:`2018`, names_to = "YEAR", values_to = "Value")
```

```
WDI_tidy <- WDI_tidy |>
  mutate(YEAR = as.integer(YEAR))

is.integer(WDI_tidy$YEAR)
```

```
[1] TRUE
```

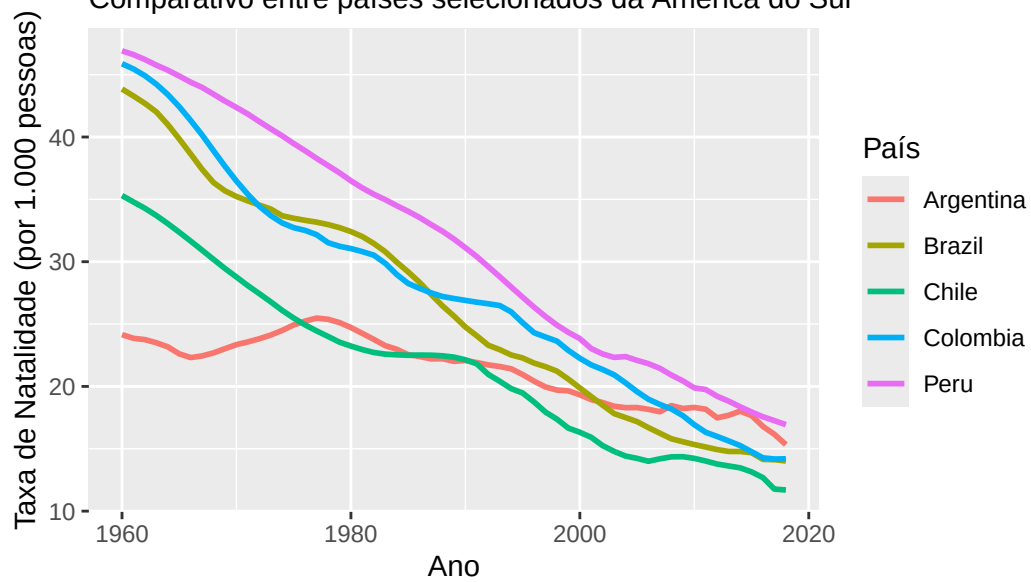
```
WDI_filtrado <- WDI_tidy |>
  semi_join(countries, by = "Country Code") |>
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") |>
  inner_join(countries |> select(`Country Code`, `Short Name`), by = "Country Code") |>
  select(`Country Code`, `Short Name`, YEAR, Value) |>
  rename(SP.DYN.CBRT.IN = Value)
```

```
WDI_ordenado <- WDI_filtrado |>
  arrange(`Country Code`, YEAR)
```

```
ggplot(WDI_ordenado, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line(linewidth = 1) +
  labs(title = "Taxa Bruta de Natalidade (1960-2018)",
  subtitle = "Comparativo entre países selecionados da América do Sul",
  x = "Ano", y = "Taxa de Natalidade (por 1.000 pessoas)", color = "País")
```

Taxa Bruta de Natalidade (1960–2018)

Comparativo entre países selecionados da América do Sul



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

[1] "2026-09-09 18:21:38 -03"